

A model of cortical associative memory based on a horizontal network of connected columns

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Abstract. An attractor network model of cortical associative memory functions has been constructed and simulated. By replacing the single cell as the functional unit by multiple cells in cortical columns connected by long-range fibres, the model is improved in terms of correspondence with cortical connectivity. The connectivity is improved, since the original dense and symmetric connectivity of a standard recurrent network becomes sparse and asymmetric at the cell-to-cell level. Our simulations show that this kind of network, with model neurons of the Hodgkin–Huxley type arranged in columns, can operate as an associative memory in much the same way as previous models having simpler connectivity. The network shows attractor-like behaviour and performs the standard assembly operations despite differences in the dynamics introduced by the more detailed cell model and network structure. Furthermore, the model has become sufficiently detailed to allow evaluation against electrophysiological and anatomical observations. For instance, cell activities comply with experimental findings and reaction times are within biological and psychological ranges. By introducing a scaling model we demonstrate that a network approaching experimentally reported neuron numbers and synaptic distributions also could work like the model studied here.

1. Introduction

1.1. Background

In this paper we present a model of the general auto-associative memory properties that the neocortex is presumed to support. One class of models used to describe this cortical property is the recurrent artificial neural network (ANN) model. In our model the functional units corresponding to the units in an ANN are represented by lamina II/III cells in cortical columns. This brings the connectivity closer to that of the cortex, since it becomes sparse, especially for the long-range connections, and also asymmetric at the level of single cells. The cell model is relatively detailed, with several compartments and ion channel dynamics of Hodgkin–Huxley type. The synapses are of several different types and have different cellular placement.

The recurrent attractor ANN model (Anderson *et al* 1977, Hopfield 1982, Cohen and Grossberg 1983, Ackley *et al* 1985, Lansner and Ekeberg 1985, Smolensky 1986, Lansner and Ekeberg 1989, Amit *et al* 1990) shares phenomenology with the associative memory function of the cortex as revealed by psychological memory experiments. For instance, in priming experiments, if subjects are first shown a list of words and then asked to fill

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in missing letters to form a word from a non-unique cue such as _ _rse (horse, curse, ...), it is more likely that they will fill in with one word from the list than one not in the list. This capability of filling in missing parts of the cue to form the complete target is very similar to the pattern completion capability of recurrent attractor networks[†]. The ANN models also display several features that may be described in gestalt psychological terms. As already mentioned above, when a partial pattern is presented, the full pattern is retrieved. Also, if two patterns are presented simultaneously, a rivalry process leads to a competition between the patterns where one pattern typically defeats the other. Phenomena such as perceptual grouping and figure-ground segregation are discussed for a recurrent network of visual integration by Tononi *et al* (1992). Furthermore, recurrent ANN models of recognition and recall of memorized words are capable of handling non-trivial exceptions, i.e. recognition failure of recallable words (Sikström and Lansner 1995). That model shows a quantitatively good agreement with experimental data for non-exceptions as well as for free recall exceptions and cued recognition exceptions. The model was also quite insensitive to catastrophic interference (pre-existing associations that could interfere with the experiment).

Recurrent attractor ANNs have also been used to model mental disturbances. The general problem of categorization, i.e. the grouping of objects into the same class and the separation between different classes, has been used as a model of prosopagnosia (a loss of being able to identify an individual object of a class) (Virasoro 1989). Further, in the study by Hoffman and McGlashan (1993), the breakdown of a unitary memory output, which was replaced by a multitude of separately active patterns, was likened to the schizophrenic loose associations, and the more-or-less autonomous groups were likened to parasitic foci. In a study by Hinton *et al* (1993), an attractor model of language with semantic and phonologic representations of written words showed reductions in functionality similar to dyslexic persons when damaged.

Several relatively detailed modelling studies focused on neocortical function are based on recurrent connectivity and obtain their principal function therefrom (Douglas *et al* 1989, Douglas and Martin 1991, Tononi *et al* 1992, Douglas *et al* 1995, Somers *et al* 1995, Suarez *et al* 1995, Hansel and Sompolinsky 1996, Amit and Brunel 1997, 1998). In models of hippocampus a recurrent scheme is also very common. Many of the former studies model parts of the visual system (Douglas and Martin 1991, Tononi *et al* 1992, Somers *et al* 1995, Suarez *et al* 1995, Hansel and Sompolinsky 1996), and investigate the performance of the model on well studied phenomena such as orientation selectivity.

On the other hand, the biological plausibility of the Hopfield network, which is the prototypical attractor network, has been questioned on several points, e.g., low storage capacity[‡], slow convergence time and high firing rates. However, improved models (typically with sparse activity) can store many memories and computer simulations done by ourselves and others have demonstrated that both objections regarding convergence time and firing rate can be overcome (Lansner and Fransén 1992, Fransén *et al* 1993, Fransén and Lansner 1995b). A remaining drawback, however, of the ANN models has been their lack of a biologically reasonable connectivity in that they are usually based on an all-to-all and symmetric[§] connectivity between neurons. The real cortical connectivity pattern is much more complex, extremely sparse (at least in general) and is also asymmetric on a cell-to-cell basis. It also shows considerable locality in that most connections are between

[†] We will use the terms assembly and pattern interchangeably throughout this paper.

[‡] For the standard Hopfield-type network the number of patterns possible to store is only $0.14N$ for patterns of size $0.5N$.

[§] For networks with both excitatory and inhibitory cells the network is mirror symmetric if one considers the RS-FS pairs, which means that the two corresponding synapses of two pairs have equal strength.

nearby cells, and long-range connections become progressively more infrequent. Due to the columnar arrangement of the neurons, the model presented here does not have this all-to-all and symmetric connectivity between neurons. Additionally, if the attractor network hypothesis is to be accepted it must be possible to give a more detailed model, with wider experimental compliance, and still have the basic functionality preserved. The aim of this work is to show that such a detailed network model can operate as an associative memory in much the same way as an attractor ANN. This has been shown for an earlier model using realistic neurons but with a simple connectivity (Lansner and Fransén 1992, Fransén and Lansner 1995a).

Our column model includes only the lamina II/III pyramidal cells and inhibitory interneurons, and the afference from lamina IV. It thus corresponds primarily to the horizontal network connecting cells in the different columns, and the local network within lamina II/III. From experimental observations the pyramidal cells are said to be dominated by input from lateral connections, and not by afference (Martin 1988). Further, most horizontal connections go between lamina II/III pyramidal cells (Gilbert and Kelly 1975, McGuire *et al* 1991). For the associative memory function studied here other afference is assumed to merely play a modulatory role. Although different neocortical areas show functional and structural specializations we have assumed they all share some basic properties in their associative memory function. Our model has therefore not been given any of the special properties of a specific area. The cells in each column constitute a functional unit, analogous to a unit defined by the cells in an orientation column, and correspond to a single unit in the ANN. As mentioned, this columnar organization brings the connectivity closer to that of real cortex, since it becomes sparse, especially for the long-range connections, and also asymmetric at the level of single cells. By introducing a set of scaling equations we also want to make it plausible that a network approaching experimentally reported neuron numbers and synaptic distributions could also work like the network studied here. This also allows for a more elaborate comparison of model numbers with anatomically determined quantities. Preliminary results on this columnar-based model have been presented previously (Lansner and Fransén 1995).

An advantage of using a relatively detailed cell model in the investigation of whether the attractor network is an adequate model of cortical associative memory function is that more comparisons may be made with experimental data. One source of single-cell activities from awake and behaving monkeys are the studies on short-term (Fuster and Jervey 1982, Funahashi *et al* 1989, Miyashita and Chang 1988) and long-term (Miyashita 1988) memory tasks which have revealed elevated firing rates for cells in the prefrontal cortex during the memory-demanding delay period of match-to-sample experiments. Much of this work is reviewed by Fuster (1995) and Goldman-Rakic (1995). In the latter study it is reported that *in vivo* experiments show that closely adjacent pyramidal and non-pyramidal neurons with similar preferences show opposite activity. When the one cell increases its firing rate the other decreases its rate (Wilson *et al* 1994, Goldman-Rakic 1995). This is different from what is found in the primary visual cortex where suppression of activity is strongest for an optimal stimulus orientation (Douglas *et al* 1991), i.e. excitation and inhibition seem to work in the same direction. Further, in the review by Goldman-Rakic (1995) it is hypothesized that cells with the same preference are excitatorily connected both locally (within a column) and long-range, and that cells with opposite or dissimilar preference provide mutually inhibitory action on each other. This mutual inhibition between columns coding for different orientations has not been shown in visual cortex where connections between iso-orientation columns seem to dominate. The elevated rates and opposed excitatory and inhibitory activity pattern of prefrontal neurons is consistent with

what is produced in our network. An auto-associative memory based on an attractor network produces elevated activity for the cells in the retrieved memory. If the learning rule that produced the connectivity is of a correlation-based (Hebbian) type, cells active in the same memory can become connected by excitatory synapses and cells active in different memories become inhibitorily connected. In our columnar network this makes nearby excitatory and inhibitory cells display opposite activity changes.

1.2. Outline of the model

In this study we have used the general purpose simulators, SWIM and SPLIT, intended for numerical simulation of networks of biologically realistic model neurons (Ekeberg *et al* 1991, 1994, Hammarlund and Ekeberg 1996). Details of the mathematical model of cells and synaptic transmission can be found in Ekeberg *et al* (1991), Brodin *et al* (1991), Wallén *et al* (1992), Tråvén *et al* (1993) and Fransén and Lansner (1995b).

The cell models used in the simulations described here are of intermediate complexity. They are of the regularly spiking pyramidal cell type (RS) with six compartments, or the non-adapting fast-spiking inhibitory type (FS) with three compartments. Voltage-dependent ion channels for Na, K, Ca, and Ca-dependent K are modelled using Hodgkin–Huxley-like equations. The excitatory RS–RS glutamate synapse is of a mixed kainate/ α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) and N-methyl-D-aspartate (NMDA) type. The RS–FS synapse has only a fast kainate/AMPA component. The inhibitory FS–RS

NETWORK WITH CELL ASSEMBLY

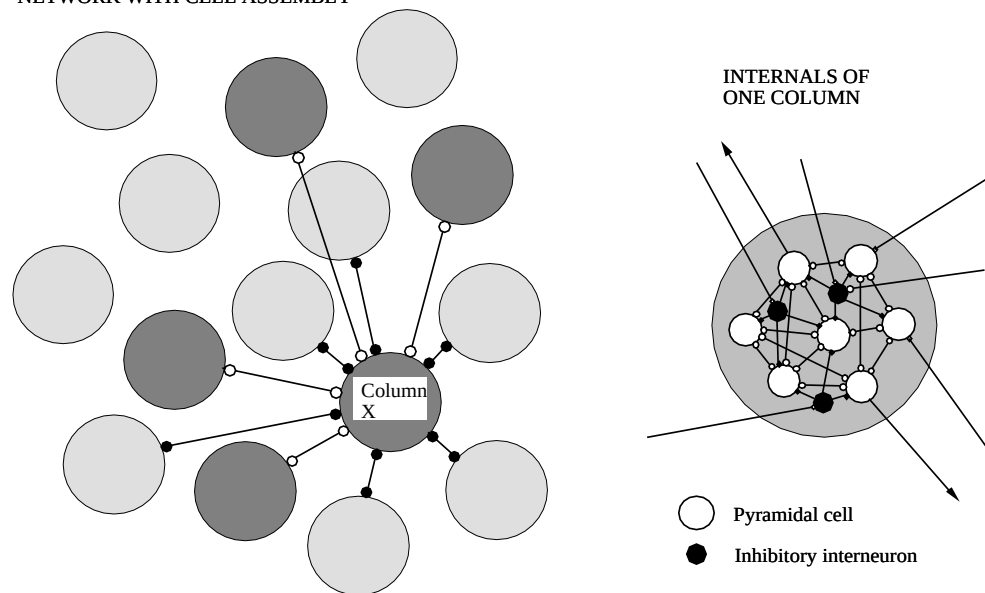


Figure 1. Left, part of the model network shown for comparison with the corresponding attractor ANN. Each filled circle is a column/network unit. Darker columns are in the same cell assembly. The connectivity of 'column X' is shown. Each intracolumnar connection shown represents multiple synaptic connections. Right, internal structure of one column comprised of excitatory pyramidal cells (white) and local inhibitory interneurons (dark). Incoming excitatory connections make contact directly onto pyramidal cells whereas inhibition is disynaptic as seen from a pyramidal cell.

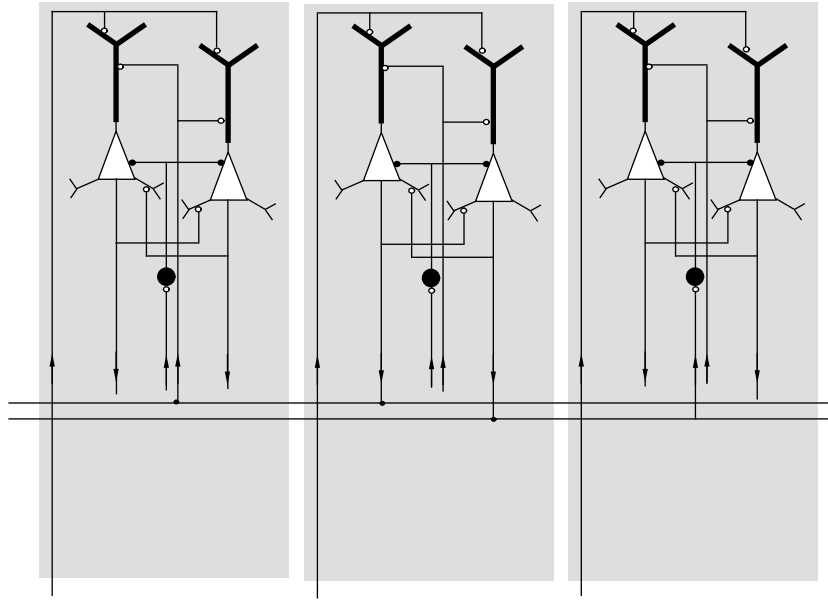


Figure 2. Wiring diagram (lateral view) for the network model simulated. Three columns are shown; the two leftmost ones belong to the same assembly.

synapse is of the fast GABA_A type. There are two calcium pools, one with a faster time constant (Ca-channel related) and one with a slower time constant (NMDA-channel related).

The network consists of 750 cells arranged in 50 columns. Each column is composed of 12 pyramidal cells and three fast-spiking inhibitory interneurons, see figures 1 and 2. Within a column the pyramidal cells connect to each other densely. The inhibitory interneurons only contact the local pyramidal cells. The excitatory long-range (inter-columnar) connectivity is from pyramidal cells in the sending column to pyramidal cells in the receiving column. Disynaptic inhibitory drive is from pyramidal cells in the sending column to the inhibitory interneurons in the receiving column. Thus local connectivity is dense and both excitatory and inhibitory, whereas the long-range (inter-columnar) connectivity is sparse and exclusively excitatory. Further, although the functional (effective) connectivity between columns is all-to-all and symmetric, the cell-to-cell connectivity is strongly asymmetric and sparse. The connectivity which determines the long-range synaptic efficacy was created using an ANN and then incorporated into our network model. The memories that are recalled in the present simulations are the assemblies/patterns that have been stored in the artificial neural network by the learning rule of the ANN. The model will be described in detail in the next three sections.

2. Structure of the network model

The basic network consists of 50 columns connected by horizontal long-range connections. Larger examples will be discussed in section 7 on scaling. The connections between the columns were created using an ANN and then incorporated into our network model. Eight different assemblies/patterns, each composed of eight out of 50 units/columns active, were stored in the ANN network by a Bayesian correlation-based (Hebbian) learning rule (Lansner and Ekeberg 1989, Lansner and Holst 1996), see appendix A. This gave 392 positive

connections and 1232 negative connections, resulting in a total of 1960 and 7392 inter-columnar excitatory synapses to pyramidal cells and inhibitory cells respectively, as will be described below. By this use of a learning rule not only excitation but also inhibition is structured, as opposed to models using global unstructured inhibition (i.e. Douglas *et al* 1995, Hansel and Sompolinsky 1996). Patterns are overlapping, i.e. sharing some common columns. Connections become excitatory between columns that occur together in one or more of the stored patterns, and inhibitory between those columns that never occur together in any of the patterns. This is consistent with a hypothesis of prefrontal cortex which proposes that cells with the same preference are excitatorily connected and cells with opposite (or dissimilar) preference provide mutually inhibitory action on each other (Goldman-Rakic 1995). Further, in prefrontal cortex it has been shown that closely adjacent pyramidal and non-pyramidal neurons show opposite activity. When the one cell increases its firing rate the other decreases its rate (Goldman-Rakic 1995). This kind of activity also appears in our network as will be shown in subsection 6.3.

Each column is composed of 12 pyramidal cells and three fast-spiking inhibitory interneurons. This represents a sub-sampling in terms of cell numbers, but gives the 80% excitatory and 20% inhibitory neurons reported by Gilbert *et al* (1990) and Katz and Callaway (1992). Within a column the pyramidal cells connect to each other at the basal dendrite by a total of 70 randomly chosen synapses. A basal placement of local excitatory synapses is described by Nicoll and Blakemore (1993). The latency (conduction and synaptic delay) of the connection is 1.2 ms as found by Mason *et al* (1991) with a standard deviation of 2.5%. Within the column our inhibitory interneurons each make a contact to the soma of eight pyramidal cells. Thus every pyramidal cell receives input from two inhibitory cells. The latency is the same as for the RS–RS connection. In cortex, a lamina II/III basket cell usually connects to pyramidal cells in its own column (Martin 1988). The basket cell connects on the soma or proximal dendrites of the pyramidal cell (Nicoll 1994, Buhl *et al* 1994). A pyramidal cell is contacted by several (10) basket cells and every basket cell contacts many (300) pyramids (Martin 1988).

In the model, many long-range connections (one per presynaptic cell) converge on the apical dendrite of the RS cell. Apical placement of long-range excitatory synapses is reported by Gilbert *et al* (1990). A very sparse (presumed singular) connectivity on vertically oriented dendrites (presumed apical) has been found (McGuire *et al* 1991). Here, the RS–RS excitatory long-range connectivity between two columns is from two presynaptic pyramidal cells in the sending column to five postsynaptic pyramidal cells in the receiving column (two and three synapses per sending cell respectively). Long-range connections onto the inhibitory cells go from six presynaptic pyramidal cells in the sending column to the soma of the three inhibitory interneurons in the receiving column (one synapse per sending cell). This will give the fast rise time of the EPSP as reported for the corresponding synapses (Thomson and Deuchars 1994), due to a proximal or somatic placement. Each model pyramidal cell has one connection on the inhibitory interneuron as found by Buhl *et al* (1994), and several cooperating inputs are needed to fire a cell (Somogyi and Martin 1985).

The modelled columns were placed on a rectangular grid with 1 mm separation representing a 7 mm × 7 mm patch of cortex. The long-range connections have a latency which varies with the ‘distance’ between columns. With an axonal conduction velocity of 1 m s⁻¹ (corresponding to unmyelinated intracortical fibres) the latency was 2.2–11 ms with an average of 5.0 ms. With myelinated fibres the distance between the columns can be much larger and still yield a latency of the same order of magnitude. A standard deviation of 2.5% was used when calculating each delay.

Thus, in the model the local connectivity is dense and both excitatory and inhibitory, whereas the long-range (inter-columnar) connectivity is sparse and exclusively excitatory. Further, although the functional (effective) connectivity between columns is all-to-all and symmetric, the cell-to-cell connectivity is strongly asymmetric and sparse. Partial support for recurrent (all-to-all) connectivity comes from the observation that long-range connections between areas are reciprocal and to multiple columns (Rockel *et al* 1980, Gilbert *et al* 1990). Also reciprocal connections between multiply interconnected regions are reported (Goldman-Rakic 1988, Cavada and Goldman-Rakic 1989). Further, in V1 symmetric columnar interactions may be present as exclusively iso-orientation columns are connected (Gilbert *et al* 1990, Ts'o *et al* 1986). A symmetric scheme may also be a result of the hypothesis by Goldman-Rakic (1995) as mentioned previously.

In section 7 a few resulting numbers (totals and proportions of synapses of different types) are compared with experimental findings and related to scaling issues and model extensions.

3. The cell and synapse models

3.1. The neocortical pyramidal cell model

Our model of the neocortical pyramidal cell of lamina II/III (Fransén and Lansner 1995b), RS, is based on the excitatory regularly spiking type cell (McCormick *et al* 1985, Connors and Gutnick 1990). A short stimulation of the model neurons gives a relatively wide spike followed by an early and a late afterhyperpolarization (AHP) separated by a small depolarizing afterpotential. At longer stimulations the cell has a pronounced adaptation. The cell is modelled with six compartments (soma, initial segment, basal compartment and three serial apical compartments). The soma diameter is sampled from a normal distribution with a mean of 21 μm and a standard deviation of 5%. The dendritic tree has one apical and one basal branch separating synaptic input of different origin. Parameter values used here are found in the appendix C and were discussed previously (Fransén and Lansner 1995b).

3.2. The fast-spiking inhibitory neuron

For the fast-spiking inhibitory interneuron, FS (McCormick *et al* 1985, Connors and Gutnick 1990), many fewer data are available. Essentially it is a non-adapting, fast-spiking small cell with a small dendritic tree. It is likely to correspond to the inhibitory cell connecting to the lamina II/III pyramids and belonging to the histologically defined basket cell type (Goldman-Rakic 1995, Martin 1988, McGuire *et al* 1991, Kawaguchi 1995). The action potential repolarization of the modelled FS cell is fast as reported by several investigators (McCormick *et al* 1985, Connors and Gutnick 1990, Baranyi *et al* 1993, Foehring *et al* 1991) and the half amplitude width is 0.3 ms (McCormick *et al* 1985, Baranyi *et al* 1993). It has a large fast AHP as commonly found (McCormick *et al* 1985, Connors and Gutnick 1990, Baranyi *et al* 1993, Foehring *et al* 1991, Kawaguchi 1995) with a depth of about 10 mV and half amplitude width of 8.0 ms, but virtually no slow AHP (McCormick *et al* 1985, Connors and Gutnick 1990, Baranyi *et al* 1993, Foehring *et al* 1991). The cell is modelled with three compartments (soma, initial segment and dendrite). The soma diameter is sampled from a normal distribution with a mean of 7 μm (McCormick *et al* 1985, Foehring *et al* 1991) and a standard deviation of 5%. The membrane time constant is 8.7 ms as described by Kawaguchi (1995), which is in between the 7.2 ms (Baranyi *et al* 1993) and 11.9 ms (McCormick *et al* 1985) found in other work. The cell has a rather

linear I - f curve up to some 200 Hz, and the primary I - f slope is much steeper than for the RS cell, which is compatible with experimental data (McCormick *et al* 1985, Foehring *et al* 1991). Parameter values and further discussion of them are found in appendix C.

3.3. The model synapses

The generic synapse model used here is described by Brodin *et al* (1991) and Tråvén *et al* (1993). The excitatory RS–RS glutamate synapse is of a mixed kainate/AMPA and NMDA type with equally large peak amplitude postsynaptic potentials (PSP) for each type alone. In our simulations the NMDA conductance has been scaled up a factor of 7 compared with the AMPA conductance to obtain this relationship. Experimentally both the local (columnar) excitatory connections and the long-range connections are reported to have mixed responses (Thomson and Deuchars 1994). The model RS–FS synapse has only a fast kainate/AMPA component, following Thomson and Deuchars (1994). The inhibitory FS–RS synapse is assumed to be of the GABA_A type as found experimentally (Thomson and Deuchars 1994). This is also consistent with reports that stimulation originating remotely, for example not in the home column, gives an essentially pure GABA_A response (Hirsch and Gilbert 1991, Kawaguchi 1995). The model synapses are saturating corresponding to a full saturation of the postsynaptic receptor pool by one presynaptic activation (Fransén and Lansner 1995b). With these saturating synapses the spike rates are relatively low also when an assembly is fully active (Fransén and Lansner 1995a, b). All synaptic conductances were sampled from a normal distribution with a standard deviation of 20%. The local RS–RS synaptic conductance is 0.38 nS and gives a soma EPSP of 0.81 mV. In experiments the EPSP amplitude of nearby cells has been reported to have a large variation with an average of 0.55 mV (Mason *et al* 1991). Several active local inputs have been estimated to be necessary to activate a RS cell (Nicoll and Blakemore 1993), which is also true for our model. Our long-range RS–RS synaptic conductance is on average 1.4 nS (0.077–2.5 nS) which gives an EPSP of 2.5 mV. In cortex the synaptic strength of the long-range connections has been estimated to be one order of magnitude weaker than the local (Gilbert *et al* 1990). Here the small number of afferent synapses (3.3 per cell) gives an artificially high value, see however the discussion in section 7. The long-range RS–FS synaptic conductance of the model is on average 0.072 nS (0.0071–0.073 nS) which gives an EPSP of 2.2 mV. Experimentally, the pyramid to inhibitory interneuron strength has been reported to be variable with a mean of 0.66 mV in the range 0.31–0.92 mV (Buhl *et al* 1994). Our local inhibitory synaptic conductance is 10.0 nS and gives an IPSP of 5.4 mV. In cortex the inhibition on a RS cell from one basket cell is mediated by some 8–10 synapses (Martin 1988) and is relatively effective (Thomson and Deuchars 1994).

4. Background activity and input to the network

In vivo recordings reveal that some cells have a higher background activity than others. This is commonly called background noise. Intracellular recordings also show that resting potentials differ. We hypothesize that this might have a functional origin. One could interpret this as a way of controlling the general firing likelihood of a cell. For the same synaptic efficacy a cell held close to the threshold is easily activated and a cell far from its threshold requires more input (higher frequency and/or more fibres active) to fire. Background activity in recurrent networks has been analysed and simulated (Amit and Brunel 1997, 1998). These studies focus on requirements for stability of (low-rate) background activity and also show how stable memories may coexist. We will look at how

differential firing thresholds and synaptic efficacies may relate to the statistics of the patterns in the network and how this gives rise to background noise. If the synaptic efficacies are tuned by the Bayesian correlation-based learning rule used here (Lansner and Ekeberg 1989) (see also appendix A), cells that take part in many patterns and hence are less discriminative and weaker indicators for any particular pattern can be connected by weaker synapses. At the same time they achieve a higher bias (lower threshold) which allows the cell to be active despite the small synaptic strengths. In our network we have implemented the bias that decreases the threshold as a constant current injection. *In vivo*, the bias could come from, for instance, auto-regulated leakage channels. Having a bias in the presence of noise would make cells with a smaller threshold more active. Thus, without direct input, the *a priori* probability of firing should be higher for a cell that is part of many patterns than for a cell that is part of just a few.

The simple model of the input from lamina IV to a column is a 50 ms current injection on eight out of 12 RS cells. To avoid common-input synchronization artifacts, current stimulation start and stop times were generated from a normal distribution with a standard deviation of 5 ms for each cell. All currents strengths were generated with a standard deviation of 10%. The stimulation of a pattern of columns has two components; for details see appendix A. If the columns belong to the stimulated pattern they are given a depolarizing current injection. If they are not part of the pattern they are given a hyperpolarizing current injection. This is a way to tell the network what is known to be present and not present. No stimulation would correspond to a 'don't know' case, and in this case the cell stays at its *a priori* firing frequency. The depolarizing component could in cortex come from the lamina IV granule cells. The hyperpolarizing component could be supplied by the lamina IV feed-forward inhibitory cells (Martin 1988, Kisvarday *et al* 1989, Somogyi *et al* 1983). Depolarizing and hyperpolarizing currents are consistent with the finding in V1 that stimulation of the thalamo-cortical projection gives a monosynaptic excitation and disynaptic inhibition in lamina III (Gilbert *et al* 1990, Hirsch and Gilbert 1991, Ferster and Lindström 1983).

5. Variation of parameter values

The model obviously contains a large number of parameters. A complete study of the effect on the model when varying each of these is not within practical computational time limits. The parameters most relevant to the study are the number of connection types, the number of connections of each type, their synaptic conductance and delay. These were varied extensively. The numbers presented here represent the biologically most relevant set found to give adequate output. Further, several parameters were randomly sampled from a distribution and thus varied from simulation to simulation. Regarding parameters in the cell models, the inhibitory cell was derived by the same procedure as the excitatory one as previously described in Fransén and Lansner (1995b). For the purpose of this study, these parameters were not varied (except for those given by a random distribution). Two sizes of the column model were constructed and simulated, i.e. the basic 50-column example and the 169-column example. The simulations were made on a SUN SPARCstation 5 and an IBM SP2 system both running UNIX. A time step of 50 μ s was used and a simulation of 500 ms took about 3 minutes of computing time on four SP2 Thin-II (RS/6000) nodes. All simulations were run a number of times with different random numbers, testing all runs for sensitivity on the parameter values with a standard deviation.

6. Simulation results

Here we will first demonstrate that a column is capable of acting as a functional unit. We will then show that an assembly of columns can give after-activity and pattern completion, and thereafter show that competition between assemblies is operating properly. The qualitative behaviour of these assembly operations is similar to that produced by a recurrent ANN (Lansner and Ekeberg 1989). The assembly operations have previously been shown to work for a smaller simplified network of connected pairs with one pyramidal cell and one fast-spiking cell (Lansner and Fransén 1992).

6.1. One column

Our studies show that a single column can act as one functional unit analogous to an ANN-unit, as also reported in Fransén and Lansner (1995b). As an example, if eight of the 12 pyramidal cells in the same column are stimulated they can activate their four non-stimulated companions in some 20–40 ms (figure 3). If four cells are stimulated the completion is slower and weaker. If two cells are stimulated (which can be regarded as spurious activity) very little completion takes place, i.e. the column shows noise tolerance. No after-activity has been observed to occur for a single active column.

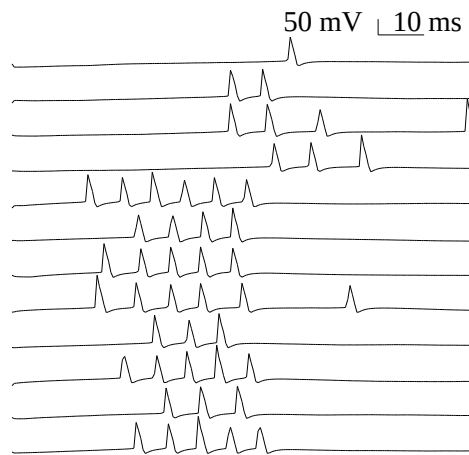


Figure 3. Completion within one column. Non-stimulated cells are activated by stimulated column members. Soma membrane potential of all 12 pyramidal cells in one column for a simulation of 100 ms and modulator application 60%. The lower eight of the pyramidal cells were stimulated with 0.2 nA for 50 ms and the remaining four were indirectly activated.

6.2. After-activity

If a complete assembly (a full pattern) is stimulated, a peak of activity occurs, see figure 4(a). The activity is terminated due to the adaptation of the cells. When the application of a neuromodulator like serotonin is simulated by decreasing the conductance of the Ca-dependent K-channel (see Fransén *et al* 1993, Fransén and Lansner 1995b and references therein) the activity in an activated assembly may instead persist, figure 4(b). The after-activity is the result of a reduced adaptation of the cells due to a decreased conductance of the Ca-dependent K-channel which, in turn, is caused by the simulated neuromodulator

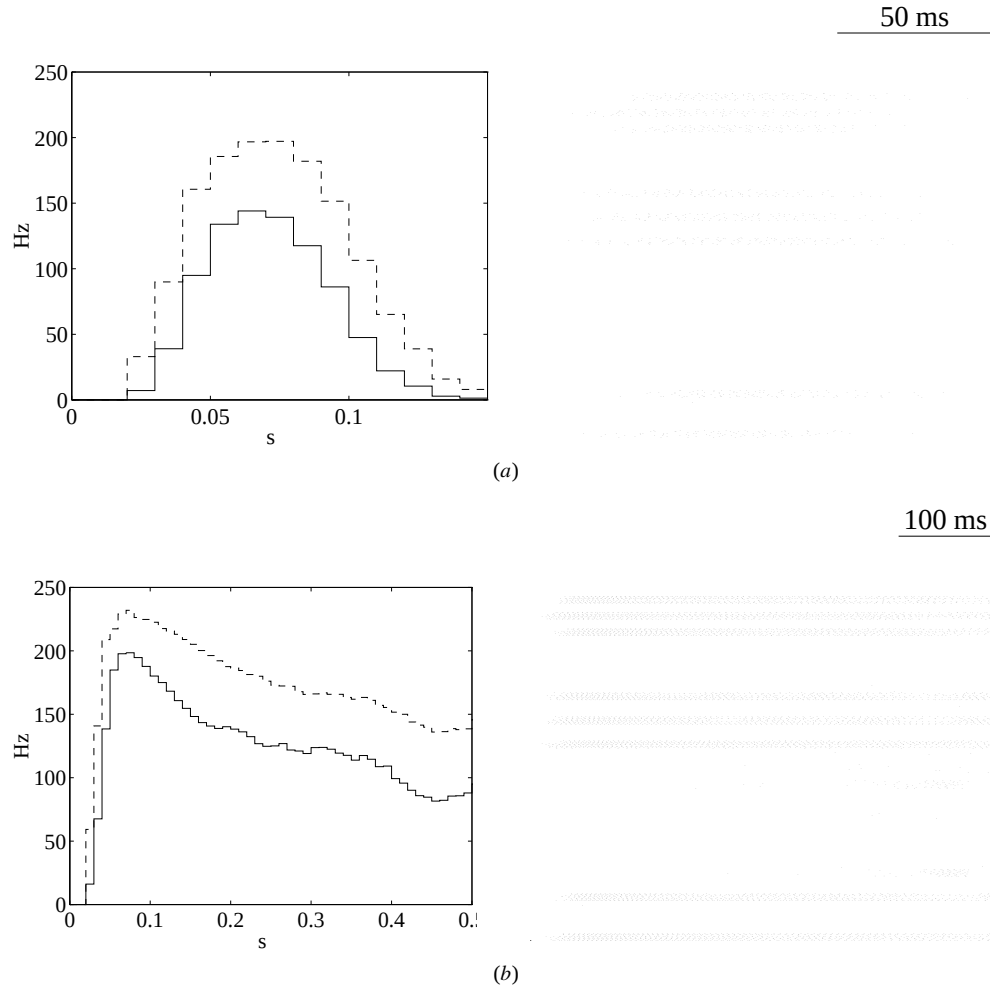


Figure 4. After-activity in one cell assembly. In the first case (a) only a transient activity peak occurs. In the second case (b) spiking activity persists over 500 ms. One full pattern is stimulated, first without simulated modulator application (a), then with 60% modulator application (b). Left, mean spike rate histogram with standard deviation added on mean, and bin width 10 ms. Right, spike raster plot showing all 600 pyramidal cells. The pattern is seen as eight groups (columns 1, 6, 25, 28, 31, 39, 41, 43), each with 12 consecutive cells. The spurious activity that can be seen (especially around 360 ms) comes from columns that do not belong to any pattern and therefore lack (inhibitory) synaptic input. Simulated time 150 ms (a) and 500 ms (b) respectively. Stimulation with 0.2 nA for 50 ms on eight of the 12 cells for each of the eight columns in the pattern.

application. The amount of modulator application used here will be given as a value between 0–100%, where 0% means no modulator present, i.e. full adaptation, and 100% means no adaptation, i.e. no Ca-dependent K-channel conductance. With modulator application all stored patterns showed stable after-activity for 400–500 ms. This shows that the stored memories can be reliably recalled. The firing of the cells in a pattern is correlated, but not on a fine (1–10 ms) time scale, as discussed in Suarez *et al* (1995) and Fransén (1996). Thereby the overall activity of the network does not depend on the exact time of a spiking event.

If the network is studied without simulated modulator application, the activity peak following the stimulation of a pattern will be followed by more peaks of activity from other patterns activated in succession. The different patterns that have been stored will in this way be activated in a non-trivial order due to their overlap. This feature is similar to that found in studies with reduced spiking models (Cartling 1995) and continuous output units (Cartling 1993). In that work this alternating activity pattern is called explorative in contrast to the case with modulator application which gives direct retrieval of one pattern.

With the summing model commonly used for synapses, the frequency in the after-activity period would have been 150–200 Hz. Here, the firing rate is 70–100 Hz which is lower than the maximum due to the saturating conductance of the synaptic model (Fransén and Lansner 1995b). The small size of the assembly is one limiting factor when looking at low rates. With a larger assembly the rate can be closer to the 20–60 Hz reported from *in vivo* recordings. In previous studies a 200 pyramidal cell assembly, for instance, showed stable activity at 70 Hz (Fransén and Lansner 1995b).

6.3. Pattern completion

When, for instance, five of the eight columns in one assembly are stimulated the remaining three will become activated (pattern completion) by the inter-columnar excitatory connections. A few randomly activated columns added to the five stimulated columns will be suppressed (noise tolerance) as in figure 5. All patterns were tested with a stimulation of five columns and they all completed their missing members and gave afteractivity for 400–500 ms. With fewer number of columns stimulated, completion may become slower

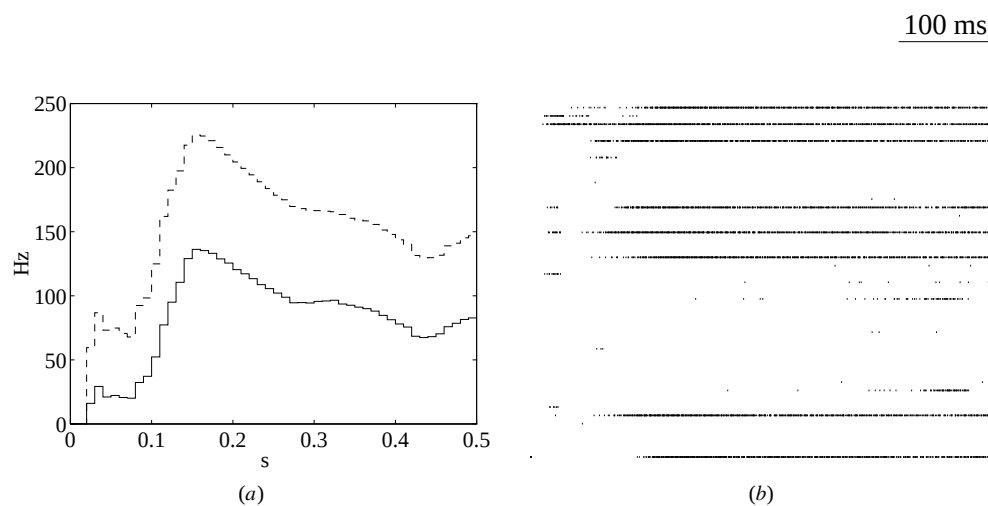


Figure 5. Pattern completion and noise tolerance. The partial and noisy stimulation leads to pattern completion of missing columns and suppression of columns which are not in the pattern. Stimulation of five columns from one pattern (the same pattern as in figure 4) and three randomly chosen columns. Left, mean spike rate histogram with standard deviation added on mean, and bin width 10 ms. Right, spike raster plot showing all 600 pyramidal cells, where all spikes from a column are put on a common line thereby showing the total column activity. The spurious activity that can be seen in the right half of the plot comes, as in figure 4, from columns that do not belong to any pattern. Simulated time 500 ms with 60% modulator application. Stimulation with 0.2 nA for 50 ms on eight of the 12 cells in a column.

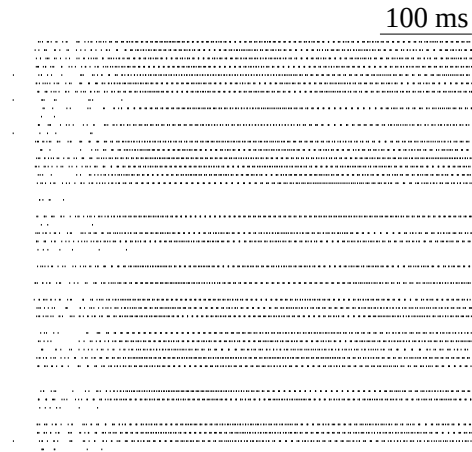


Figure 6. Spike raster plot showing all 150 inhibitory cells where all spikes from a column are put on a common line. Inhibitory inactive columns are either part of the active pattern or not part of any pattern.

and less reliable. With just one or two stimulated columns the recalled pattern (if any) is not well defined. Further, the inhibitory cell activity is essentially the complement of the excitatory activity on the column level, see figures 5 and 6. This is analogous to findings in prefrontal cortex as discussed in subsection 1.1 and section 2.

The pattern completion and noise suppression are taking place shortly after stimulation has begun. This process is relatively fast, a complete pattern is normally obtained in 30–60 ms. It may thus be done within the initial activity peak without after-activity. This short reaction time can be compared with that found in psychophysical reaction time experiments (Thorpe and Imbert 1989, Thorpe *et al* 1996), as has been discussed in previous work (Lansner and Fransén 1992). Those experiments set a lower limit of 100–150 ms, measured by event-related potentials, for the identification of familiar visual objects.

6.4. Pattern rivalry

An overlap between assemblies, i.e. cells participating in more than one assembly, is essential for having an acceptable storage capacity[†]. Despite this overlap, an activated assembly will not produce any spurious secondary activation of columns in other assemblies. The spread of activation is prohibited by the lateral inhibition between assemblies. This pattern separation capability is essential for a storage system. However, if the overlap is too large the assemblies cannot be separated but will be grouped (activated) together. In our network the overlap between the eight patterns with eight columns each is 0–3 columns. When parts of two assemblies are stimulated (conflicting input) they will compete. When one of the assemblies wins it will complete its pattern and shut down the activity in the other one, see figures 7 and 8. This could be a physiological correlate of perceptual rivalry effects. After-activity may here give the network more time to settle down in a stable complete activity pattern. Note in the figure that the firing pattern of individual cells varies considerably from rather regular to irregular (bursting). This is a network property as the isolated cell fires regularly.

[†] Without this overlap a network of size N and assembly size m could only store N/m assemblies.

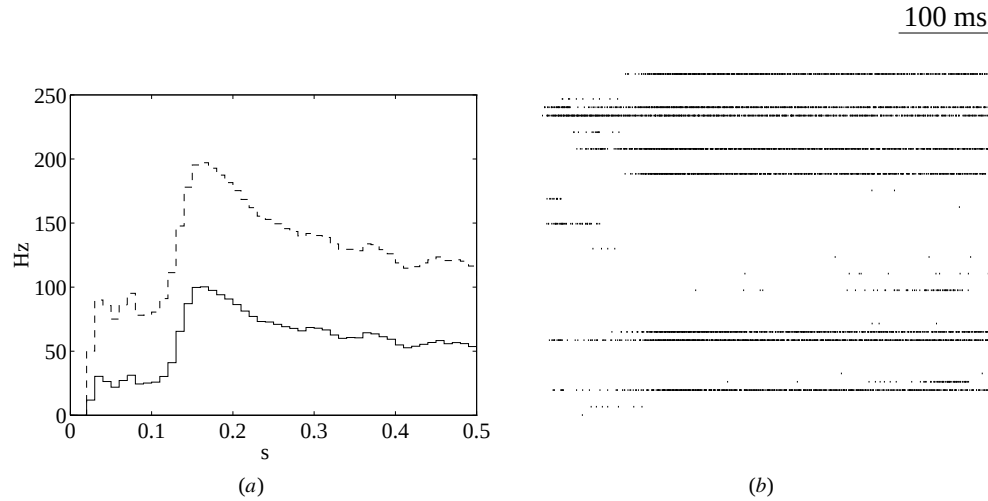


Figure 7. Pattern rivalry. The simultaneous stimulation of two partial patterns leads to a rivalry process whereby one pattern defeats the other. Stimulation on four columns from one pattern (columns 28, 31, 41, 43) and four columns in another pattern (columns 8, 14, 41, 42). Left, mean spike rate histogram with standard deviation added on mean, and bin width 10 ms. Right, spike raster plot showing all 600 pyramidal cells where all spikes from a column are put on a common line. The spurious activity that can be seen in the right half of the plot comes, as in figure 4, from columns that do not belong to any pattern. Simulated time 500 ms with 60% modulator application. Stimulation with 0.2 nA for 50 ms on eight of the 12 cells in a column.

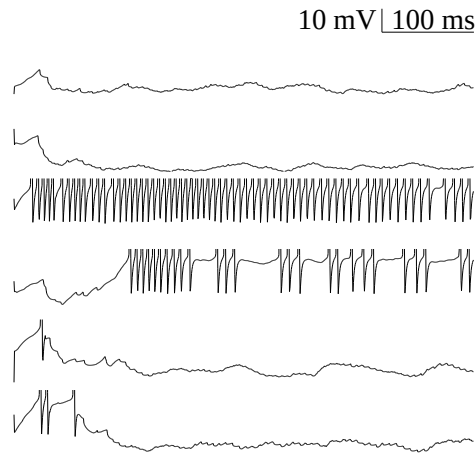


Figure 8. Pattern rivalry. Sample soma membrane potential for two cells in the winning pattern (middle two), two cells in the defeated pattern (lower two) and two cells of columns belonging to neither pattern (upper two). Action potentials are truncated. Note in the figure that the firing pattern of individual cells varies considerably from rather regular to irregular (bursting). After-activity may here give the network more time to settle down in a stable complete activity pattern.

7. Scaling of cell numbers and synaptic distributions

The network we have simulated is very small which has forced us to make compromises. For instance, in order to obtain a sufficiently high number of EPSPs per unit time onto

the FS cells in order to activate them properly, the number of RS–FS synapses has been exaggerated. In the first part of this section we will look at numbers and distributions of cells and synapses in the network and relate them to experimental findings. It is also important to show that the network performance is not degraded when its size is increased and the relative proportions of connections are changed in the direction of more realistic values. In the latter part of this section we will study a scaling model and give an example using this.

7.1. Discussion of the connectivity parameters in the simulated model

In total our basic 50-column network has 12 852 excitatory synapses and 1200 inhibitory synapses. In cortex the basket cell gives some eight synapses on a target pyramidal cell (Martin 1988). This gives a multiplicative factor of 8 for the inhibitory synapses in the model when their number is compared with the excitatory synapses. For the basic model this yields 43% inhibitory synapses[†]. This can be compared with the estimates of 11% reported for inhibitory synapses (Schüz and Palm 1989). Our relatively large number of inhibitory synapses was a consequence of the requirement of a sufficient number of IPSPs per unit time on an inhibited cell. For a somewhat larger simulated network of 169 columns trained with 21 patterns with 14 columns per pattern, the proportion was 21%. In the scaling example below we could adhere to the experimental value.

Concerning the density of connections, about half of the pyramidal cells have been estimated to send out lateral connections (Cynader *et al* 1988). This figure is used in our scaling model and roughly realized in our basic model. Further, we have assumed that they all send local connections. As the local volume is much smaller than the volume restricted by the lateral connections (extending up to some 8 mm), local connectivity becomes relatively dense and long-range very sparse (we will also return to this in a paragraph below). In cortex the proportion of inhibitory cells is only 20%, and furthermore the inhibitory cells have a much smaller membrane area. In our basic network the inhibitory cell has 5.4 times more input synapses on its dendrites compared with an excitatory cell. The denser connectivity is in agreement with the observation that inhibitory cells have about one order of magnitude larger synaptic density than pyramidal cells (McGuire *et al* 1991).

It has been reported that the local connection probability between pyramidal cells is in the order of 9% (Mason *et al* 1991, Nicoll and Blakemore 1993). In the basic example the local synapses give a larger probability, 53%, because the number of cells per column is so small. For more realistic numbers see the discussion on local and long-range connections below. Two cells are quite rarely mutually connected (Mason *et al* 1991). In our 50-column network the probability of a reciprocally connected pair in a column is 28% and between columns it is 0.5%. The intracolumn probability is again high due to the small number of cells per column.

7.2. Background to the scaling model

Generally, we have used a ‘functional’ strategy when looking how to define our scaling equations. By this we mean for instance observing how the number of active synapses on a cell scales. On average in the basic example an RS cell gets 5.8 synapses from its column companions and 1.5 synapses from RS cells belonging to the other seven columns of an active pattern. As mentioned before, these numbers are small as the network only represents

[†] The number of inhibitory synapses multiplied by eight and divided by the sum of this and the number of excitatory synapses.

a sub-sampling. The relative proportion, however, is satisfactory as it allows a cell to be activated by local as well as distant inputs.

Regarding the origin of synapses, 72% has been estimated to be local excitatory connections (relative all excitatory connections) within a radius of 300 μm and 28% outside (Nicoll and Blakemore 1993). For smaller radii, like the one of our columns, the proportion of local connections is most likely smaller. If we assume that there is a Gaussian connection probability we obtain 1.3% inside our column, which will be used in the scaling model. For the basic 50-column network we have 27% intracolumnar connections as too few local synapses per cell otherwise would prevent the cells within the column from being sufficiently correlated.

As for the number of cells in a column, the number of neurons underneath a square millimeter cortex has been estimated to be 146 000 (Rockel *et al* 1980) (except for primary visual cortex where it is about 2.5 times greater). A column is here estimated to be 50 μm $\times$ 50 μm . This estimate is taken from the 800–1000 μm found for the distance between ocular dominance columns (Löwel 1994), each housing some 18 iso-orientation columns, and the distance of 56 μm between clusters of apical dendrites (Peters and Yilmaz 1993). Based on the number of neurons per square millimetre and the size of a column, the number of neurons in an orientation column can be estimated to be some 900. Other areas than V1 thus would have some 350 neurons if the same area is assumed. As the lamina II/III cells constitute some third of these (Peters and Yilmaz 1993) below we will use 120 cells as our estimate of the number of lamina II/III RS cells and 30 for the FS cells.

7.3. The scaling model

To study how the network might be scaled up towards more realistic sizes, we have set up a number of scaling equations for different quantities. The equations are found in appendix B and their predictions will be illustrated here by an example. As a starting point we take the upper 8 mm length of a long-range connection that has been reported. We use this length as the diagonal in a square piece of cortex which then gets a side of 5.7 mm. For this area we obtain 12 800 columns of side 50 μm . For the proportion of local synapses we take our estimate of 1.3% and for the proportion of inhibitory synapses we take the 11% that was experimentally reported. From the discussion of cell numbers above we will assume we have 120 RS cells and 30 FS cells per column, giving a total of 1.9 million cells. In earlier studies we have shown that the increased size of the column will have a beneficial effect, an increased number of cells per pattern will also give decreased firing rates (Fransén and Lansner 1995b). Further, we will assume an average activity level of 1% of the cells which gives us a pattern size of 128. As an estimate of the number of active long-range synapses onto a cell in a pattern we have chosen 10, based on previous experiences with the 50-column network. This is probably at the lower end, but one should also remember that this network is only a part of the whole network and thus cannot account for all the synaptic connectivity. As this value is some seven times that for the 50-column network, the PSPs could here be correspondingly smaller. A detailed account of the calculations can be found in appendix B and a summary of some of the numbers in table 1.

Assuming these values the scaling model gives the following. We obtain 1.55 billion long-range synapses to RS cells and 0.39 billion long-range synapses on the FS cells. The former value gives some 1008 synapses per receiving RS cell. From the latter number we obtain 1027 synapses per receiving FS cell and 257 RS–FS synapses per RS cell. These numbers are still a bit small but become adequate if larger patches are considered. Also, this is as already mentioned only a part of the cortical network. For instance, we have

Table 1. Summary of relative proportions.

Quantity	50 columns	169 columns	Scaling example	Cortex
Per cent inhibitory synapses	43	21	11	11
Per cent local connections	27	11	1.3	72 (1.3 ^a)
Local connection probability	0.53	0.53	0.14	0.09

^a Our estimate.

not included the afference from lamina IV and other vertical circuitry. The 2528 synapses we obtain for a sending RS cell is quite reasonable. For the local connection probability our derived value of 0.14 is only out by a factor of 1.6, which is quite satisfactory. The value for the number of (local) synapses per FS cell gives some 2850 synapses on the 120 RS cells which should be enough to control the column activity. Finally, the value for the number of local synapses per RS cell gives 17 synapses per cell. The number of active long-range synapses per cell was set to 10, so both types should have about equal possibility of affecting the cell.

8. Discussion

8.1. Model assumptions

We have used a columnar organization of the cells in our network model. A modular description of cortex has been put forward by Mountcastle (1957), Hubel and Wiesel (1962), and Szentágothai (1978). Anatomic evidence for a patchy structure has been obtained by, for instance, a combination of 2-deoxyglucose autoradiography and retrograde labelling (Gilbert *et al* 1990, McGuire *et al* 1991). A functionally patchy structure has been shown *in vivo* by cross-correlation analysis (Hata *et al* 1993, Ts'o *et al* 1986) and *in vitro* by a multi-site stimulation technique combined with whole-cell patch clamp for recording (Weliky and Katz 1994). Weliky and Katz report strong correlations between nearby cells, which is consistent with our column model where several cells have similar afference and recruit each other in order to operate as one functional unit. The view of a patchy (columnar) structure of the neocortex has support not only in primary sensory areas (orientation columns, ocular dominance columns, colour blobs, vibrissa barrels) but is also found in association areas, i.e. entorhinal cortex (Hevner 1993), as discussed by Hevner and Wong-Riley (1992).

Our network has a *functionally* full and symmetric connectivity between columns. Partial anatomical support for this type of region or area connectivity comes from the observation that long-range connections between areas are reciprocal and to multiple columns (Rockel *et al* 1980, Gilbert *et al* 1990). Also reciprocal connections between multiply interconnected regions are reported (Goldman-Rakic 1988, Cavada and Goldman-Rakic 1989). In that work the authors use the term ‘multiple nodal links’. These links connect regions by multiple and usually reciprocal connections. The total effect of one column on another has however not been determined experimentally to our knowledge. The rare instances examined experimentally where two cells have mutual connections indicate that the synaptic efficacies are not of equal size (Mason *et al* 1991), i.e. there is asymmetric connectivity at the cell level when two cells are mutually connected. We have not tested our model for this case beyond the asymmetry given by the standard deviation of the synaptic conductance (20%). Furthermore it is not known what type of circuit these

synapses belong to. A network supporting sequential association, for instance, as opposed to the auto-association studied in this report, would be expected to have an asymmetric connectivity.

We have not constrained the model to any specific cortical area. A pattern (active cells in a certain set of columns) in our network can be viewed as a cell assembly. Indeed we like to follow the ideas of Hebb and others that the cell assembly may be dispersed over several cortical areas. Support for this view is, for instance, presented and discussed by Fuster (1995). A specialization would thus not be necessary for the basic function. This also results in a processing that is commonly called parallel, as opposed to serial or feed-forward. The experimental support for parallel and serial processing is discussed by, e.g., Bullier and Nowak (1995). The main conclusion is that although anatomical studies favour a serial interpretation, functional studies give this very little support. The order of activation of areas in the visual system, for instance, does not follow that suggested by the anatomical (serial) analysis, and there seems to be a good deal of simultaneity in the activity. There is also structural support for the ideas based on distributed and recurrent processing, e.g., Goldman-Rakic (1988) and Cavada and Goldman-Rakic (1989), as mentioned previously.

Why is experimental detection of assembly activity so hard? Hebb suggested that the cell assembly was the basic representational unit. We have assumed that the assembly has a distributed representation which can possibly extend over large cortical regions. With a low activity level, such an entity will be difficult to detect experimentally, see also the discussion about firing rates and activity in Fransén and Lansner (1995b). One further explanation why assemblies have not been found so far can be taken from the suggestions made by Cynader *et al* (1988) that computational maps representing 'information-bearing parameters' need to have little or no topographic order. The analysis of experimental criteria for detection of assemblies by Strangman (1996) shows that detecting an assembly may indeed be hard, but not beyond practical limits.

8.2. Model limitations

In our model we have obviously done several simplifications. For instance, the cortex has six lamina with feed-forward and recurrent connections between them. Our column model includes only the lamina II/III pyramidal cells and inhibitory interneurons, and the afference from lamina IV. Further, there is only one type of interneuron included and there is no inhibition on these. The network thus corresponds primarily to the horizontal network connecting cells in the different columns, and the local network within lamina II/III. For the fundamental cortical memory functions we have studied, we have assumed that, except for the input from lamina IV, the influence in lamina II/III from the vertical connections is weak. From experimental observations the pyramidal cells are said to be dominated by input from lateral connections, and not by afference (Martin 1988). Further, most horizontal connections go between lamina II/III pyramidal cells (Gilbert and Kelly 1975, McGuire *et al* 1991). This lamina is also especially thick in the frontal and parieto-temporal 'association' areas (Rockel *et al* 1980). The activity of lamina IV, the major afference (Gilbert and Wiesel 1979, Kisvarday *et al* 1989, Ferster and Lindström 1985, Blasdel *et al* 1985), is here modelled by the stimulation supplied to the neurons. Other afference, the minor direct thalamocortical input (Ferster and Lindström 1983) as well as the small input (excitatory and inhibitory) from lamina V and VI (Kisvarday *et al* 1989, Ferster and Lindström 1983, Somogyi *et al* 1983, Blasdel *et al* 1985) have been omitted. For the associative memory function studied here they are assumed to merely play a modulatory role. Given

this, the lamina II/III, together with its horizontal connections and input from lamina IV, may be studied separately.

8.3. Using a relatively detailed model

With a relatively detailed model more extensive comparisons against data from electrophysiological experiments can be made. In the review by Goldman-Rakic (1995) a qualitative model is presented where cells with the same preference are excitatorily connected both locally (within a column) and long-range, and where cells with opposite or dissimilar preference provide mutually inhibitory action on each other. This is consistent with our model. With a quantitative model as ours, we have further been able to make several comparisons both on connectivity numbers and proportions, and on cellular activities. For instance, among the values the scaling model predicts, the local connection probability and the number of axonal terminals per neuron differ from experimental values only by some factor 2. Moreover, the cells in an active pattern show elevated firing rates and excitatory cells change firing rates differently from inhibitory ones as reported in prefrontal cortex.

The increased model complexity gives the model some properties distinct from those of traditional ANNs. During the after-activity period we obtain a semi-stable activity which to some degree resembles that of a limit cycle. It has, however, the long-time stability of an attractor state. In figure 8 we can see that the firing pattern of individual cells of our simulations also varies considerably from rather regular to irregular (bursting). It is due to a network property as an isolated model neuron fires regularly. This can be compared to many cortical registrations revealing a large variability in cell firing patterns. Furthermore, the adaptation of the RS cell (which may be affected by modulators) gives the network some new properties different from standard ANNs with non-adapting cells. The adaptation gives an intrinsic mechanism to end an active pattern so that a new one may be activated. Also, the saturating synaptic model gives sub-maximal firing rates (Fransén and Lansner 1995b) as opposed to what is produced in a standard ANN.

8.4. The scaling model

The purpose of the scaling example was to show how our 50- and 169-column networks might be scaled up towards more realistic sizes and cell densities. Due to the small size of the 50-column network, and to a lesser extent the 169-column network, there are deviations from both what is experimentally found and from the scaling model. One common adjustment, see, e.g., Fransén and Lansner (1995b) and Amit and Brunel (1998), is the compensation for the otherwise too small number of incoming PSPs per unit time. Here, for instance, the number of RS cells sending long-range synapses to FS cells and the number of synapses between two columns to FS cells are both larger than expected. This is a compensation for the fact that when numbers become very small, large variations in number of inputs or number of PSPs per unit time might otherwise appear. Further, the scaling model does not allow for an arbitrarily large number of columns, as discussed in appendix B. For large numbers, additional factors such as sparsity in the connectivity between columns would have to be included. However, a larger pattern can most likely tolerate not being all-to-all even between columns, and still function as an entity. The increase may even be beneficial. When the number of columns per pattern increases one may achieve increased robustness for pattern completion, as was noted for the 169-column example. Further, the network does not need to be limited to the dramatically sub-sampled 5.7 mm $\times$ 5.7 mm given in

the example. A sub-sampling of columns over a larger area, where the inter-column fibres are myelinated, would be equivalent. In the scaling example the pattern size of 1% is a factor of five higher than the optimum (Lansner and Ekeberg 1985), and will give better redundancy at a price of lower storage capacity. For the selected activity level we will thus have 1% activity among the RS cells. We also obtain activity in most FS cells of columns which are correlated with but do not belong to the active pattern, yielding a much higher activity level among this class of inhibitory neurons.

8.5. Computational aspects of the model

This model has several computationally favourable features. The attractor dynamics gives noise tolerance and pattern completion. This kind of operation seems to be present in PRS (priming (Tulving and Schacter 1990)) as discussed in the subsection 1.1. Pattern completion and noise tolerance were also observed in the model of orientation selectivity by Douglas *et al* (1995). As patterns are overlapping the storage capacity is satisfactory. This is important as the storage capacity of the cortex appears to be quite large. The pattern completion and noise suppression are taking place shortly after stimulation has begun. This process is relatively fast, a complete pattern is normally obtained in 30–60 ms. It may thus be done within the initial activity peak without after-activity. As discussed in subsection 6.3 and in a previous work (Lansner and Fransén 1992), this short reaction time can be compared with that found in psychophysical reaction time experiments (Thorpe and Imbert 1989, Thorpe *et al* 1996). Those experiments set a lower limit of 100–150 ms, measured by event-related potentials, for the identification of familiar visual objects. Short reaction times were also produced by a recurrent network used to model integration in the visual system (Tononi *et al* 1992). The recurrency, of intra- as well as inter-area connections, of that network was shown to be crucial for the model to perform form from motion and motion capture. Further, the powerful signal processing capabilities displayed by a recurrent network has been discussed in relation to cortical processing (Somers *et al* 1995, Suarez *et al* 1995).

Many relatively detailed neocortical model studies represent parts of the visual system (Douglas and Martin 1991, Tononi *et al* 1992, Somers *et al* 1995, Suarez *et al* 1995, Hansel and Sompolinsky 1996), and investigate the performance of the model on experimentally well studied phenomena such as orientation selectivity. By studying a special area, i.e. V1, and investigating features related to the visual process, these models will inevitably have differences, as also discussed by Douglas *et al* (1995), compared with ours modelling the general auto-associative memory function of the cortex. The former models may display a proportional behaviour with respect to an input parameter such as contrast or velocity. A memory model, on the other hand, displays one out of the many stored patterns upon presentation of a stimulus (and thus responds highly non-linearly with respect to its input).

8.6. Model predictions and suggestions

A key feature of this model is the assumption that a cortical column is the basic computational unit. A possible explanation, suggested by our model, for the existence of functionally similar cells in cortical columns is that this duplication of cells is necessary in order to support a large enough number of connections to other parts of the cortex.

In vivo recordings reveal that some cells have a higher background activity than others. This is commonly called background noise. Intracellular recordings also show that resting potentials differ. We hypothesize that this might have a functional origin. One could interpret this as a way to control a cell's general firing likelihood. Having a bias in the presence of noise would make cells with a smaller threshold more active. Without direct input, as a consequence of our Bayesian learning rule, the *a priori* probability of firing should be higher for a cell that is part of many patterns than for a cell that is part of just a few.

The network has an inherent dynamics which favours the activity of a single pattern before a mixture of patterns. This can be seen as a kind of gestalt phenomenon. When parts of two assemblies/patterns are stimulated (conflicting input) they will compete. When one of the assemblies wins it will complete its pattern and shut down the activity in the other one, see figures 7 and 8. This could be a physiological correlate of perceptual rivalry, i.e. as experienced when viewing the ambiguous Necker cube picture.

9. Conclusions

Attractor network models of cortical associative memory functions have developed considerably over the past few years. Here we show that we can improve them further, in terms of correspondence with cortical connectivity, by replacing the single cell as the functional unit by multiple cells in cortical columns connected by long-range fibres. The connectivity is improved, since the original dense and symmetric connectivity of a standard recurrent network now is sparse and asymmetric at the cell-to-cell level. Our simulations show that this kind of network, with model neurons of the Hodgkin–Huxley type arranged in columns, can operate as an associative memory in much the same way as previous models having a simpler connectivity. The network showed attractor-like behaviour and could perform the standard assembly operations despite differences in the dynamics introduced by the more detailed cell model and network structure. Further, the model has become sufficiently detailed as to allow for evaluation against electrophysiological and anatomical observations. For instance, cell activities comply with experimental findings and reaction times are within biological and psychological ranges. By introducing a scaling model we demonstrate that a network approaching experimentally reported neuron numbers and synaptic distributions also could work like the model studied here. As the basic connectivity comes from an ANN, the computational features of the model may also be efficiently studied and evaluated.

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Appendix A. The ANN model

The Bayesian neural network model (Lansner and Ekeberg 1989) is designed to calculate the posterior probabilities of some output attributes given some input attributes. In the network there is one unit for each input attribute x_i and each output attribute y_j . If Bayes' rule for

conditional probabilities is combined with an assumption of independent input attributes we obtain

$$P(y | \mathbf{x}) = P(y) \frac{P(\mathbf{x} | y)}{P(\mathbf{x})} = P(y) \frac{P(x_1 | y)}{P(x_1)} \frac{P(x_2 | y)}{P(x_2)} \dots \frac{P(x_n | y)}{P(x_n)}. \quad (A1)$$

When taking the logarithm (base e) of (A1), the right-hand side becomes a sum. In this sum each term (except the first) corresponds to the weight w_{ji} from unit x_i to unit y_j :

$$w_{ji} = \ln \left(\frac{P(x_i | y_j)}{P(x_i)} \right)$$

and the bias β_j can be identified as the first term:

$$\beta_j = \ln(P(y_j)). \quad (A2)$$

Each output unit sums its inputs and exponentiates the result, yielding the desired probability. If the independence assumption is relaxed a higher order network results (Lansner and Holst 1996, Holst 1997b).

In the ANN used there is no difference between input and output units, so all units receive inputs from all others as well as external input. The dependence of the storage capacity and optimal pattern size on network size is known from simulations (Lansner and Ekeberg 1985, 1989):

$$\begin{aligned} m &= k_m \ln N & k_m &\approx 2.7 \\ Z_m &= \frac{k_z N^2}{\ln^2 N} & k_z &\approx 0.14. \end{aligned}$$

We set the current for the *a priori* activity according to $I_\beta = k_\beta e^{\beta_j}$. The current for a cell which is part of the pattern is set to $I_+ = I_{\text{stim}}$, and the current for a cell which is not part of the pattern is set to $I_- = k_- e^{\beta_j} \alpha$, where k_β and k_- are scaling constants and $\alpha = 1/Z$.

In the above expressions, N is the number of columns, m the number of columns per pattern, n the number of pyramidal cells per column[†], Z the number of stored patterns, and Z_m the maximum number of patterns that can be stored.

According to the learning rule, correlations between patterns influence the size of the individual connections. Columns that are part of many assemblies acquire weaker connections than less frequent columns. When calculating the synaptic conductances, the weight is multiplied by one scaling constant for the positive weights and one for the negative weights. Connections with approximately zero conductance were removed.

Appendix B. Scaling equations for the network

The following equations define the scaling model of which the 50-column network is the starting point. The notation is as defined in appendix A.

The number of RS–RS synapses within the column is

$$s_{\text{le}} = k_e n \quad k_e \leq n \quad n \gg 1. \quad (B1)$$

The number of FS–RS synapses within the column[‡] is

$$s_{\text{li}} = k_i 0.25n \quad k_i \leq 8n. \quad (B2)$$

[†] The number of inhibitory cells is $0.25n$.

[‡] In cortex 80% excitatory and 20% inhibitory neurons are reported by Gilbert *et al* (1990) and Katz and Callaway (1992) and the basket cell is shown to give some eight synapses on a target pyramidal cell (Martin 1988).

The number of RS–RS synapses between columns[†] is

$$s_{ge} = \frac{N^2 f_e}{2} \quad N \gg 1. \quad (\text{B3})$$

The number of RS–FS synapses between columns[†] is

$$s_{gi} = \frac{N^2 f_i}{2}. \quad (\text{B4})$$

The proportion of intracolumnar excitatory synapses in relation to all excitatory synapses (we use equations (B1), (B3), (B4)) is

$$p_e = \frac{N s_{le}}{N s_{le} + s_{ge} + s_{gi}} = \frac{1}{1 + N(f_e + f_i)/2nk_e}. \quad (\text{B5})$$

The proportion of (intracolumnar) inhibitory synapses in relation to all (excitatory and inhibitory) synapses (we use equations (B1)–(B5)) is

$$p_i = \frac{8N s_{li}}{8N s_{li} + N s_{le} + s_{ge} + s_{gi}} = \frac{1}{1 + k_e/2k_i p_e}. \quad (\text{B6})$$

The number of active intercolumnar synapses per RS cell is

$$s_a = \frac{m-1}{N} \frac{s_{ge}}{Nn} = \frac{(m-1)f_e}{2n} \quad (\text{B7})$$

$$\Rightarrow f_e = \frac{2s_a n}{m-1}. \quad (\text{B8})$$

The proportion of intercolumnar sending pyramidal cells in relation to all pyramidal cells[‡] is

$$o_e + o_i = 0.5n \quad (\text{B9})$$

$$\Rightarrow o_i = 0.5n - o_e. \quad (\text{B10})$$

The proportion of FS cells that are connected by one RS cell in relation to all (RS and FS) cells connected by the RS cell[§] is

$$\frac{N f_i}{2} = 0.25 \left(\frac{N f_e}{2} + n k_e \right) \quad (\text{B11})$$

$$\Rightarrow f_i = 0.25 \left(f_e + \frac{2n k_e}{N} \right). \quad (\text{B12})$$

Equations (B12), (B5)

$$\Rightarrow k_e = \frac{1.25 f_e}{2n/N(1/p_e - 1.25)} \quad (\text{B13})$$

and (B6)

$$\Rightarrow k_i = \frac{k_e}{2p_e(1/p_i - 1)}. \quad (\text{B14})$$

[†] For a reasonably well filled network the number of connections to columns, with RS cells or FS cells as targets, are equal and non-zero (Holst 1997a).

[‡] Roughly half of the pyramidal cells have been estimated to send out lateral connections (Cynader *et al* 1988).

[§] Pyramidal cells are reported to have 81% pyramidal cells and 19% inhibitory cells as targets (McGuire *et al* 1991).

The total number of connections in s_{ge} and s_{gi} is

$$f o_e = \frac{N^2 f_e}{2N} \quad f o_i = \frac{N^2 f_i}{2N} \quad (B15)$$

$$\Rightarrow \quad f = \frac{N f_e}{2 o_e} = \frac{N f_i}{2 o_i}. \quad (B16)$$

The two expressions in (B15) are equal, so

$$\frac{o_e}{f_e} = \frac{o_i}{f_i}. \quad (B17)$$

Equations (B10) and (B12)

$$\Rightarrow \quad o_e = \frac{1}{2.5/n + k_e/N f_e}. \quad (B18)$$

The connection probability between RS cells in a column is

$$p_c = \frac{s_{le}/n}{n-1} = \frac{k_e}{n-1} \approx \frac{k_e}{n}. \quad (B19)$$

In the above expressions k_e is the number of synapses to and from an RS cell, k_i the number of synapses from an FS cell, f_e the number of synapses to RS cells between two connected columns, f_i the number of synapses to FS cells between two connected columns, o_e the number of RS cells per column sending long-range synapses to RS cells, o_i the number of RS cells per column sending long-range synapses to FS cells, and f the number of synaptic terminals per RS cell sending long-range synapses.

As discussed in section 7 we assume $N = 12\,800$, $n = 120$, $m = 128$, $p_e = 0.013$, $p_i = 0.11$ and $s_a = 10$. From this we obtain $f_e = 19$ using equation (B8). From f_e we obtain $s_{ge} = 1.55 \times 10^9$ by using equation (B3) and $k_e = 17$ by using equation (B13). From k_e we obtain $k_i = 79$ using equation (B14), $f_i = 4.8$ using equation (B12), $o_e = 48$ using equation (B18), and $p_c = 0.14$ from equation (B19). With f_i in turn we obtain $s_{gi} = 0.39 \times 10^9$ from equation (B4). Lastly, with o_e we obtain $f = 2528$ from equation (B16) and $o_i = 12$ from equation (B10). Our scaling equations do not allow for an arbitrarily large number of columns N . The major problem is the quadratic dependence on N of s_{ge} and s_{gi} . For larger spatial distances in cortex it is reasonable to assume that the number of connections grows as something in between N and N^2 . By introducing a constant multiplicative factor less than 1 to s_{ge} and s_{gi} , much larger N may be possible.

Appendix C. Model parameters

Table C1. Parameters describing the ion channel kinetics. The notation corresponds to that used in Ekeberg *et al* (1991). The parameter values are according to Fransén and Lansner (1995b).

		Na ⁺		K ⁺	Ca ²⁺	NMDA
		<i>m</i>	<i>h</i>	<i>n</i>	<i>q</i>	<i>p</i>
α	<i>A</i> (mV ⁻¹ ms ⁻¹)	0.58	0.232	0.058	0.232	2.03 (ms ⁻¹)
	<i>B</i> (mV)	-50	-50	-50	10	—
	<i>C</i> (mV)	1	1	0.8	11	17
β	<i>A</i> (mV ⁻¹ ms ⁻¹)	0.174	1.16 (ms ⁻¹)	0.0145	0.0029	0.0292 (ms ⁻¹)
	<i>B</i> (mV)	-59	-46	-40	10	—
	<i>C</i> (mV)	20	2	0.4	0.5	17

Table C2. Parameters describing the cell. Here, m is the modulator multiplication factor. The notation in the parameter column corresponds to that used in Ekeberg *et al* (1991). The FS values are the same as in Fransén and Lansner (1995b) unless followed by a reference. The RS values are the same as in Fransén and Lansner (1995b). Only values differing from the FS ones are listed.

Parameter	FS value	RS value	Note	Unit	Description
E_{leak}	−75			mV	Leak current equilibrium potential
E_{Na}	50			mV	Sodium current equilibrium potential
E_{K}	−80			mV	Potassium current equilibrium potential
E_{Ca}	150			mV	Calcium current equilibrium potential (voltage gated)
$E_{\text{Ca(NMDA)}}$	20			mV	Calcium current equilibrium potential (NMDA gated)
g_m	0.74	0.44	a	$\mu\text{S mm}^{-2}$	Membrane leak conductance
C_m	0.01			$\mu\text{F mm}^{-2}$	Membrane capacitance
g_{ext}	0.19	0.068	b	$\mu\text{S mm}^{-2}$	Conductance from diffuse transmitter stimulation
Parameters specific to the soma compartment					
g_{Na}	150			$\mu\text{S mm}^{-2}$	Sodium conductance
g_{K}	1002	84	c	$\mu\text{S mm}^{-2}$	Potassium conductance
$g_{\text{K(Ca)}}(\text{AHP})$	0.368 M	36.8 M	d	nS	Potassium conductance (Ca^{2+} gated)
ρ_{AP}	1.0			$\text{mV}^{-1} \text{ ms}^{-1} \text{ mm}^{-2}$	Ca^{2+} inflow rate (voltage gated)
δ_{AP}	9			s^{-1}	Ca^{2+} decay rate (voltage entered Ca^{2+})
a_s	0.000 154	0.001 39	e	mm^2	Soma membrane mean area
Parameters specific to the initial segment					
g_{Na}	2505			$\mu\text{S mm}^{-2}$	Sodium conductance
g_{K}	5010	418	c	$\mu\text{S mm}^{-2}$	Potassium conductance
g_{core}	300			$\mu\text{S mm}^{-2}$	Core conductance per membrane area
a_{IS}	0.1			—	Initial segment membrane area factor (fraction of soma area)
Parameters specific to the dendrite compartments					
$g_{\text{K(Ca)}}(\text{NMDA})$	—	40 M	f	nS	Potassium conductance (NMDA- Ca^{2+} gated)
δ_{NMDA}	—	2	f	s^{-1}	Ca^{2+} decay rate (NMDA entered Ca^{2+})
g_{core}	13.7	8.1	g	$\mu\text{S mm}^{-2}$	Core conductance per membrane area
a_d	4			—	Dendrite membrane area factor (fraction of soma area)

^a The value of FS: g_m is calculated from the time constant reported in Kawaguchi (1995).

^b The value of FS: g_{ext} is adjusted to give an (*in vivo*) resting potential of −60 mV. This is 10 mV below the firing level (Baranyi *et al* 1993) which is 25 mV above the resting potential of −75 mV (Foehring *et al* 1991); see also the discussions in Baranyi *et al* (1993), Scholfield (1978), Connors *et al* (1982), Bernander *et al* (1991) and Vogt (1985) on *in vitro* and *in vivo* differences.

^c The value of FS: g_{K} has been adjusted to give a shorter spike fall time and narrower spike width (McCormick *et al* 1985, Connors and Gutnick 1990, Baranyi *et al* 1993, Foehring *et al* 1991). (The value of g_{Na} is the same as for RS (McCormick *et al* 1985, Connors and Gutnick 1990, Baranyi *et al* 1993, Foehring *et al* 1991).)

^d The value of FS: $g_{\text{K(Ca)}}(\text{AHP})$ has been set to give a barely noticeable AHP (McCormick *et al* 1985, Connors and Gutnick 1990, Lacaille 1991, Baranyi *et al* 1993, Foehring *et al* 1991).

^e The value of FS: a_s soma area from McCormick *et al* (1985) and Foehring *et al* (1991).

^f The value is the same as in Tråvén *et al* (1993).

^g The value of FS: g_{core} is calculated from a length constant of 0.233 per compartment. This is the same as for the RS cell. See also the discussion on page 462 of Shepherd and Koch (1990) on the ever shorter electrotonic lengths used in simulations.

Table C3. Parameters describing the synapse. The notation used in the parameter column corresponds to that used in Tråvén *et al* (1993). The data in the values column are the same as in Fransén and Lansner (1995b) unless followed by a reference. Key: C, connection within the column; L, long-range connection between columns; I, connection to a FS cell.

Parameter	Value	Note	Units	Description
Parameters specific to the kainate/AMPA synapse				
Compartment	Basal (C), medial (L), somatic (I)	a, b	—	Location on postsynaptic target
E_{Na+K}	0		mV	Equilibrium potential for Na^+/K^+ current
g_{exc}	0.38 (C), 1.4 (L), 0.072 (I)	c	nS	Conductance after spike
t_{delay}	1		ms	Time delay
$t_{duration}$	0		ms	Open time
ρ_{syn}	0		ms	Raise time constant
δ_{syn}	10		ms	Closing (decay) time constant
Parameters specific to the GABA _A synapse				
Compartment	Soma	d	—	Location on postsynaptic target
E_{Cl}	−85	e	mV	Equilibrium potential for Cl^- current
g_{inh}	10.0	c	nS	Conductance after spike
t_{delay}	1	e	ms	Time delay
$t_{duration}$	0	e	ms	Open time
ρ_{syn}	0	e	ms	Raise time constant
δ_{syn}	17	e	ms	Closing (decay) time constant
Parameters specific to the NMDA synapse				
Compartment	Basal (C), medial (L)	a	—	Location on postsynaptic target
E_{NMDA}	0		mV	Equilibrium potential for Na^+/K^+ current
$E_{Ca(NMDA)}$	20		mV	Equilibrium potential for Ca^{2+} current
g_{NMDA}	2.7 (C), 9.6 (L)	c	nS	Conductance after spike
ρ_{NMDA}	$1.1g_{NMDA}$	e	$s^{-1} mV^{-1} \mu S^{-1}$	Ca^{2+} inflow rate (NMDA gated)
t_{delay}	5		ms	Time delay
$t_{duration}$	20		ms	Open time
ρ_{syn}	5		ms	Raise time constant
δ_{syn}	150		ms	Closing (decay) time constant
Parameters specific to the noise synapse				
Compartment	Proximal		—	Location on cell
E_{noise}	0		mV	Equilibrium potential for noise synapse
g_{noise}	105 (RS), 23 (FS)	f	pS mm^{-2}	Conductance of noise synapse
δ_{syn}	10		ms	Closing (decay) time constant

^a The combined kainate/AMPA and NMDA synaptic connection between RS cells has been placed on the basal (local within column) (Nicoll and Blakemore 1993) or medial (long-range between columns) (McGuire *et al* 1991) dendritic compartment.

^b The synaptic connection between a RS cell and a FS cell is a kainate/AMPA synapse. It has been placed on the soma compartment (Thomson and Deuchars 1994).

^c The standard deviation of the conductance of each synaptic type is 20% of its mean value. The synaptic conductances of the long-range RS–RS and RS–FS synapses come from a learning rule. The value refers to the average.

^d The placement is somatic or proximal (Nicoll 1994, Buhl *et al* 1994).

^e The value is the same as in Tråvén *et al* (1993).

^f The value of FS: g_{noise} , the conductance of the noise, is tuned to give some 1–3 spikes per 10 s as in Fransén and Lansner (1995b). As described in Wallén *et al* (1992), the source of the synaptic noise is a Poisson process with an expectation value of 300 PSPs s^{-1} .

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